

Embedded Pilot Code Aided Delay Doppler Domain Channel Estimation

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Abstract—In this paper, we propose an embedded pilot code-aided delay-Doppler (DD) domain channel estimation scheme for orthogonal time-frequency space (OTFS) modulation. At the transmitter, a code of length L with small autocorrelation sidelobes, such as the Barker code, is embedded in the DD transmit grid as a pilot sequence. At the receiver, we compute the cross-correlation of the received symbols with the known pilot sequence. Due to the ideal autocorrelation property of the pilot code, we can estimate the DD channel transfer function with high precision. This embedded pilot code allows lower peak power while maintaining a good SNR. Thus, compared to the single symbol pilot-aided estimation method, the proposed method does not incur the peak-to-average-power ratio (PAPR) penalty while achieving good performance even at low SNRs and converging to the ideal channel estimation and BER performance. The code set's low autocorrelation sidelobe peak and cross-correlation peaks ensure accurate channel estimation while limiting the guard band overhead and the PAPR.

Index Terms—OTFS, channel estimation, embedded pilot, MIMO-OTFS, Barker code.

I. INTRODUCTION

Orthogonal time-frequency space (OTFS) modulation [2] has gained significant attention due to its ability to transform the time-varying channel into a sparse, flat-fading, 2D delay-Doppler (DD) grid. This provides many advantages over orthogonal frequency division multiplexing (OFDM) systems, which perform poorly in high Doppler scenarios. To achieve reliable performance in an OTFS communication system, accurate channel state information (CSI) must be available at the receiver as a DD channel impulse response. Several pilot-aided channel estimation techniques [4], [7] - [9] have been proposed to estimate the DD channel response.

The embedded pilot-aided delay-Doppler channel estimation schemes [4], [7] utilizes a single amplified pilot symbol embedded within the data frame, separated by a guard band, as shown in Fig. 1-(a). This method provides spectral efficiency; however, the embedded pilot symbol requires additional power to perform accurate channel estimation. The higher pilot SNR results in an unfavorable peak power, as shown in Fig. 1 (c), and peak-to-average-power ratio (PAPR), inducing an additional load on the transmit amplifier. This also makes the scheme susceptible to a severe channel fade at the location of the pilot symbol and to eavesdropping schemes. To address these shortcomings, we propose an embedded pilot code, as shown in Fig. 1 (b), at the transmitter and a corresponding cross-correlation-based estimation algorithm at the receiver.

In this paper, we propose a general framework that can be extended to codesets that have the property of small autocorrelation peak sidelobes such as the Barker code, Frank codes, P1-P4 codes and Zadoff-Chu sequences. The choice of codes is made based upon the desired system specifications such as frame size, required code length, modulation scheme etc. We demonstrate the proposed approach using the Barker codes [6], which have a magnitude 1 autocorrelation peak sidelobe, making them an ideal candidate as a spreading pilot sequence. These pilot codes allows for spreading the required pilot power for accurate estimation, resulting in a lower signal peak, as shown in Fig. 1 (d), and PAPR compared to the single-symbol embedded pilot scheme.

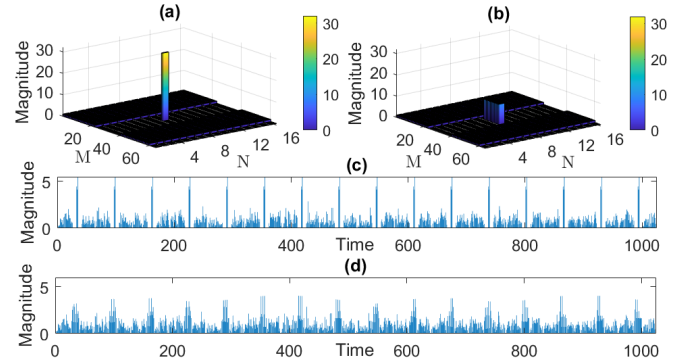


Fig. 1. (a) and (b) are the DD grid for the embedded single pilot symbol and the proposed embedded pilot code, respectively. (c) and (d) are their corresponding time domain samples.

We compare the performance of the proposed approach with the single-symbol pilot-aided estimation method under various channel conditions and demonstrate the accuracy of the channel estimation performance of the proposed method, particularly at low SNR levels.

II. OTFS SYSTEM MODEL

A delay-Doppler grid with delay resolution $1/M\Delta f$ and Doppler shifts $1/NT$ is created where M is the number of subcarriers, Δf is the subcarrier spacing in Hz, N is the number of symbols and $T = (\frac{1}{\Delta f})$ is the symbol time. The bandwidth is defined as $B = M\Delta f$. Let $X, Y \in \mathbb{C}^{M \times N}$ be the two-dimensional transmitted and received DD grids, respectively. $X_{DD}[k, l]$ represents the NM information symbols in the two-dimensional DD domain, where $k = 1, 2, \dots, N$ and $l = 1, 2, 3, \dots, M$ represent the delay and Doppler components, respectively. The information symbols are zero-padded appropriately before further processing. This

DD domain signal is transformed into the time-frequency domain signal, $X_{\text{tf}}[n, m]$, where $n = 1, 2, 3, \dots, N$ and $m = 1, 2, 3, \dots, M$ represent the time and frequency components, respectively, using the inverse symplectic fast Fourier transform (ISFFT). The signal to be transmitted, $s(t)$, is obtained by the Heisenberg transform of $X_{\text{tf}}[n, m]$. The symbols are transmitted using a rectangular pulse-shaping waveform $g_{\text{tx}}(t)$, implemented through a diagonal matrix G_{tx} containing samples of $g_{\text{tx}}(t)$.

$$s = \text{vec}(G_{\text{tx}} \cdot (X_{\text{DD}} \cdot F'_n)) \quad (1)$$

Here, F_n is an N -point discrete Fourier transform matrix and F'_n represents its inverse. At the receiver, the information symbols can be recovered from the received signal $r(t)$ by demodulating it, followed by detection. This process is accomplished by transforming back to the DD domain using the Wigner transform, resulting in $Y_{\text{tf}}[n, m]$, followed by the SFFT, resulting in $Y_{\text{DD}}[k, l]$.

$$Y_{\text{DD}} = \text{vec}^{-1}(F_n \cdot (I_M \otimes G_{\text{rx}}) \cdot r) \quad (2)$$

where I_M is an $M \times M$ identity matrix and $\otimes$ represents the Kronecker product. Symbol detection is performed on $Y_{\text{DD}}[k, l]$ to recover the transmitted information symbols. We assume that $H(\tau, \nu)$ is the complex channel impulse response. τ_{max} and ν_{max} represent the maximum delay and the maximum Doppler shift among all channel paths. Since typically there are only a small number of reflectors in the channel with associated delays and Dopplers, very few parameters are needed to model the channel in the delay-Doppler domain. The sparse representation of the channel is

$$H(\tau, \nu) = \sum_{i=1}^P H_i \delta(\tau - \tau_i) \delta(\nu - \nu_i) \quad (3)$$

Here, P is the number of propagation paths, h_i , τ_i , and ν_i represent the complex gain, delay, and Doppler shift associated with the i -th path, and $\delta(\cdot)$ denotes the Dirac delta function.

We denote by l_{τ_i} , k_{ν_i} the delay and Doppler taps for the i -th path as

$$\tau_i = \frac{l_{\tau_i}}{M\Delta f} \quad \nu_i = \frac{k_{\nu_i}}{NT} \quad (4)$$

Let us denote l_{τ} and k_{ν} the delay and Doppler taps corresponding to the largest delay τ_{max} and Doppler ν_{max} . The relation between $Y_{\text{DD}}[k, l]$ and $X_{\text{DD}}[k, l]$ was derived in [1] as where $\hat{H}[k', l'] = H[k', l'] e^{-j2\pi \frac{k'}{NT} \frac{l'}{M\Delta f}}$, $b[k', l'] \in \{0, 1\}$ is the path indicator, i.e., $b[k', l'] = 1$ indicates that there is a path with Doppler tap k' and delay tap l' with corresponding path magnitude $\hat{H}[k', l']$, otherwise, there is no such path, i.e., $b[k', l'] = 0$ and $\hat{H}[k', l'] = 0$. Finally, the term $W[k, l] \sim \mathcal{CN}(0, \sigma^2)$ is an additive white noise with variance σ^2 , and $[\cdot]_N$, $[\cdot]_M$ denote modulo N and M operations, respectively. The total number of paths is:

$$\sum_{k'=-k_{\nu}}^{k_{\nu}} \sum_{l'=0}^{l_{\tau}} b[k', l'] = P. \quad (5)$$

Each path circularly shifts the transmitted symbols by the delay and Doppler taps.

III. EMBEDDED PILOT CODE-AIDED CHANNEL ESTIMATION

We assume OTFS with ideal waveforms for a multipath channel with integer and fractional Doppler cases. Let $x_d[k, l]$ denote the data symbols located at location $[k, l]$ in the delay-Doppler information grid, and 0 denotes the guard symbol. The guard symbols ensure that the received symbols for channel estimation do not interfere with the received data symbols. The SNR for the pilot and data symbols is taken as the same. We first define the central grid location $[k_p, l_p]$ such that $k_p = N/2$ and $l_p = M/2$. The authors of [4] place a single pilot symbol X_p at (k_p, l_p) within the delay-Doppler transmit grid. The guard band is chosen such that the channel estimation can be performed for the maximum specified delay and Doppler tap values, i.e., l_{τ} and k_{ν} . The total number of guard symbols is $(2k_{\nu} + 1)(2l_{\tau} + 1) - 1$. The channel is estimated using a thresholding algorithm on the received symbols $k \in (k_p - k_{\nu}, k_p + k_{\nu})$, $l \in (l_p - l_{\tau}, l_p + l_{\tau})$. To reliably differentiate between channel taps and the noise floor, the authors of [4] add additional power of up to 40 dB to X_p . Instead of placing a single pilot symbol in the delay-Doppler transmit grid, we place the Barker code as a pilot sequence x_b in the OTFS transmit grid. The Barker code [6] is a finite-length sequence of binary digits ($\in \{+1, -1\}$) that minimizes the off-peak autocorrelation coefficients. Barker codes have magnitude 1 autocorrelation sidelobes. However, the longest known Barker code is of length 13. For our calculations, we define the Barker sequence as a vector $x_b[n]$ for $0 \leq n \leq L - 1$ where L is the length of the Barker sequence. We arrange the pilot, guard, and data symbols in the delay-Doppler grid for an OTFS frame transmission as in Fig. 2:

$$x[k, l] = \begin{cases} x_b, & k = k_p, l_p - l_g \leq l \leq l_p + l_g \\ 0, & k_p - 2k_{\nu} \leq k \leq k_p + 2k_{\nu}, \\ & l_p - l_g - l_{\tau} \leq l \leq l_p + l_g + l_{\tau}, \\ x_d[k, l], & \text{otherwise.} \end{cases} \quad (6)$$

where $l_g = (L - 1)/2$. The number of guard symbols are $N_g = (4k_{\nu} + 1)(2l_g + 2l_{\tau} + 1) - L$. At the receiver, we use the received symbols $Y_{\text{DD}}[k, l]$, $k_p - k_{\nu} \leq k \leq k_p + k_{\nu}$, $l_p - l_g \leq l \leq l_p + l_g + l_{\tau}$ for channel estimation. We can express the received symbols for channel estimation as,

$$Y_{\text{DD}}[k, l] = b[k - k_p, l - l_p] \hat{H}[k - k_p, l - l_p] x_b + W[k, l] \quad (7)$$

for $k \in [k_p - k_{\nu}, k_p + k_{\nu}]$, $l \in [l_p - l_g, l_p + l_g + l_{\tau}]$. We can see that if there is a path with Doppler tap $k - k_p$ and delay tap $l - l_p$, we have $Y_{\text{DD}}[k_p, l_i] = \hat{H}[k_p, l_i] x_b + W[k_p, l_i]$ where $l_p - l_g \leq l_i \leq l_p + l_g$. Otherwise, $Y_{\text{DD}}[k, l] = W[k, l]$. At lower SNRs, the $|\hat{H}[k_i, l_p]|$ cannot be accurately distinguished from the noise $W[k_i, l_p]$, resulting in the detection of false taps that lower the system performance. The authors of [4] utilize a higher pilot SNR to improve detection

accuracy; however, an increased PAPR makes such a scheme undesirable. We propose a cross-correlation-based channel estimation algorithm to detect the channel taps accurately.

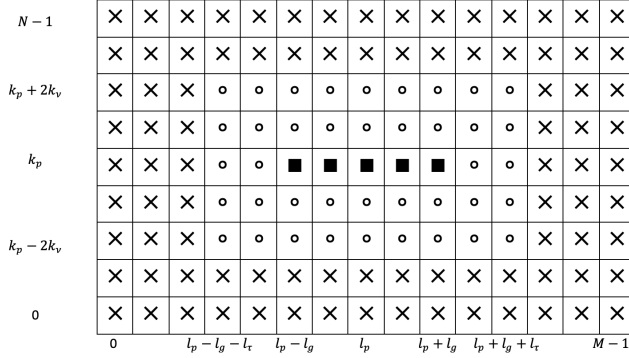


Fig. 2. Tx symbol arrangement (■: pilot sequence; o: guard symbols; x: data symbols)

First, we compute the cross-correlation of the received delay-Doppler symbols $Y_{DD}[k_i, l_i]$ with the Barker code x_b .

$$R_{y,x}[k_i, l_i + C] = \sum_{n=-l_g}^{l_g} Y_{DD}[k_i, l_i + n] x_b^*[n] \quad (8)$$

where $C = M/2 - L$. For $k_i = k - k_p$ and $l_i = l - l_p$, for $k \in [k_p - k_v, k_p + k_v]$, $l \in [l_p - l_g, l_p + l_g + l_\tau]$. Which reduces to,

$$R_{y,x}[k - k_p, l - l_p + C] = LT + \sum_{n=-l_g}^{l_g} W[k - k_p, l - l_p] x_b[n] \quad (9)$$

Since we assume that $\hat{H}[k - k_p, l - l_p] = T$. Fig. 3-(a) shows a DD domain 3GPP channel, and Fig. 3-(b), the receiver corresponding cross-correlation output. If the magnitude $|R_{y,x}[k - k_p, l - l_p + C]| \geq \tau$, where τ is some positive detection threshold, then we estimate $b[k - k_p, l - l_p] = 1$ and $\hat{H}[k - k_p, l - l_p] = R_{y,x}[k - k_p, l - l_p + C]/L$. Otherwise, we set $b[k - k_p, l - l_p] = \hat{H}[k - k_p, l - l_p] = 0$.

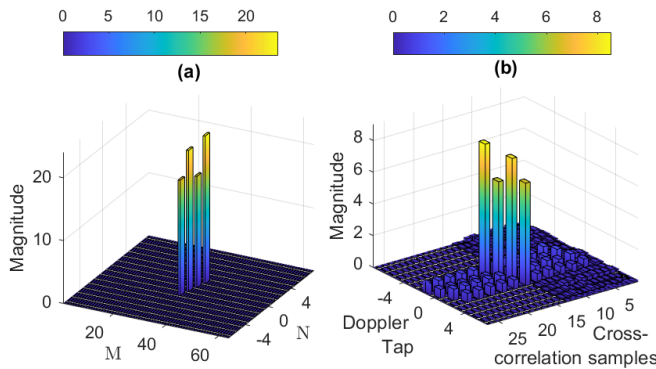


Fig. 3. (a) DD domain 3GPP channel. $taps = 4, l_\tau = 3, k_v = 2$, (b) Receiver cross-correlation output.

The proposed channel estimation algorithm is designed to improve the accuracy of the detection of taps by minimizing the detection of false taps by providing a significant difference between the magnitude of $R_{y,x}$ and the noise floor when

a path exists. We then use the estimated information for data detection, i.e., the received symbols $y[k, l]$ for data detection are

$$Y_{DD}[k, l] = \sum_{k'=-k_v}^{k_v} \sum_{l'=0}^{l_\tau} z[k', l'] + W[k, l] \quad (10)$$

where $z[k', l'] = b[k', l'] \hat{H}[k', l'] x_d[k - k', l - l']$, and $k \notin [k_p - k_v, k_p + k_v]$ or $l \notin [l_p - l_g, l_p + l_g + l_\tau]$. Note that we have a total of $MN - (2k_v + 1)(2l_g + l_\tau + 1)$ received symbols to detect a smaller number of $MN - (2l_g + 2l_\tau + 1)(4k_v + 1)$ data symbols.

IV. SIMULATION RESULTS

The BER for the proposed embedded pilot code-aided channel estimation scheme is shown in this section. The system parameters are as follows. The carrier frequency is 2 GHz, the subcarrier spacing is 15 kHz, the number of symbols per frame is $N = 16$, and the number of subcarriers $M = 128$. The bandwidth of a single frame is therefore 1.92 MHz. For our simulations, we consider a synthetic 3GPP channel consisting of 4 channel taps with the maximum delay $l_\tau = 3$ and maximum Doppler $k_v = 2$. We utilize a Barker code of length $L = 11$ to code the pilot symbols. Consequently, the overheads for the pilot and guard symbols are 0.54 % and 6.93 % respectively. We compare this with the single embedded pilot-based channel estimation scheme as proposed in [4], wherein the overheads for the pilot and guard symbols are 0.049 % and 3.03 %, respectively. Due to the reduced guard band size, the single embedded pilot scheme transmits more data symbols per frame for the given frame size. However, for our computations, we assumed the number of transmitted data symbols to be the same across both schemes to perform a fair comparison. An additional required power of 7 dB has been added to the pilot symbols to streamline the performance. At the receiver, we have used a time-domain linear minimum mean square error (LMMSE) detector for channel equalization.

The BER depends directly upon the accuracy of the channel estimation and hence is a good measure of the efficiency of the proposed channel estimation algorithms. The SNR of the single pilot symbol (SNR_{p_u}) and the SNR of the pilot code symbols (SNR_{p_c}) have been given a small additional power such that $SNR_{p_u} = SNR_{p_c} = 9$ dB. The BER vs SNR curve comparing the performance of a single pilot symbol and the Barker code-based pilot scheme is shown in Fig. 4. Here, (SNR_{p_u}) and (SNR_{p_c}) have been kept the same in both schemes. The Barker code-based pilot scheme achieves ideal channel estimation performance, while the single pilot symbol channel estimation scheme performs slightly worse. This performance improvement is a unique advantage gained through the ideal autocorrelation properties of the Barker code that can be used to localize the channel taps at the receiver under noisy conditions.

The magnitude of the peaks of the pilot symbols is equal to that of the data symbols. This helps us reduce the PAPR of the transmitted frame. The authors of [4] increase SNR_{p_u} to improve the BER performance, leading to increased PAPR.

Fig. 5 (a) represents the PAPR CCDF curve for the case when $SNR_{p_u} = SNR_{p_c}$. We can see that the PAPR CCDF curves for the two cases converge.

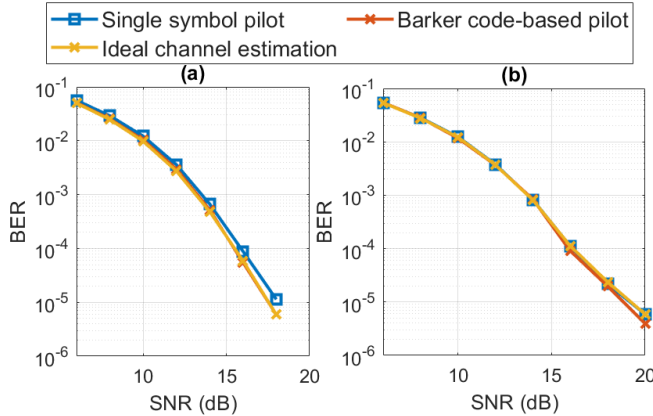


Fig. 4. BER vs SNR for (a) $SNR_{p_u} = SNR_{p_c}$, i.e., the total transmitted power is different between schemes, and (b) $SNR_{p_u} = LSNR_{p_c}$, i.e., the total transmitted power is the same between schemes. Here $N = 16, M = 128, 4 - QAM$.

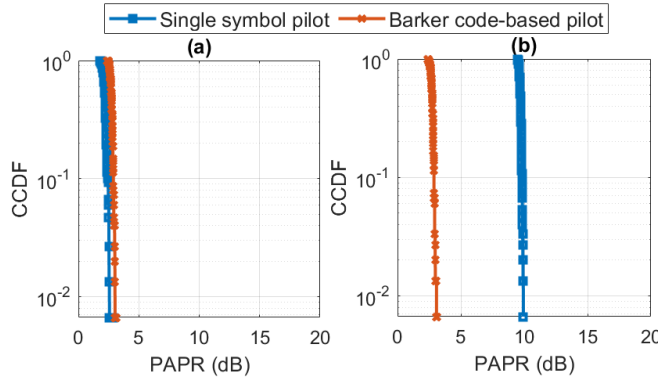


Fig. 5. PAPR CCDF for (a) $SNR_{p_u} = SNR_{p_c}$ i.e. the total transmitted power is different between schemes, and (b) $SNR_{p_u} = LSNR_{p_c}$ i.e. the total transmitted power is the same between schemes. Here $N = 16, M = 128, 4 - QAM$.

Fig. 4 (b) compares the BER performance of the single pilot symbol scheme and the Barker code-based pilot scheme for the case when $SNR_{p_u} = LSNR_{p_c}$, i.e., the total power of the transmitted frame is kept the same between the two schemes. Both perform equally well in this scenario, converging to the ideal channel estimation curve. Here, both schemes achieve a BER of 10^{-3} at just 14 dB. Fig. 5 represents the PAPR CCDF curve of the case when $SNR_{p_u} = LSNR_{p_c}$. When we compare the PAPR curves in 5 (a) and (b), a significant difference of almost 6 dB can be observed between the two schemes. Fig. 6 represents the RMSE curves for the two scenarios i.e. when (a) $SNR_{p_u} = SNR_{p_c}$ i.e. the total transmitted power is different between schemes, and (b) $SNR_{p_u} = LSNR_{p_c}$ i.e. the total transmitted power is the same between schemes. The coded channel estimation scheme has a lower RMS error and thus allows for more accurate channel estimation. Simulation results highlight the effectiveness of the proposed pilot code-aided channel estimation scheme that achieves good BER performance without incurring a large PAPR penalty.

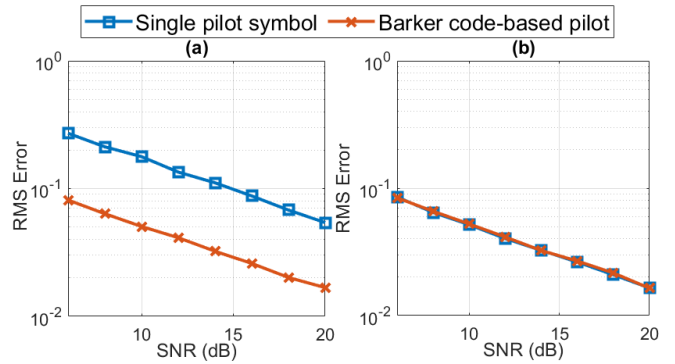


Fig. 6. RMSE curves for (a) $SNR_{p_u} = SNR_{p_c}$ i.e. the total transmitted power is different between schemes, and (b) $SNR_{p_u} = LSNR_{p_c}$ i.e. the total transmitted power is the same between schemes. Here $N = 16, M = 128, 4 - QAM$.

V. CONCLUSION

In this paper, we propose an embedded pilot code-based channel estimation scheme for OTFS. Barker codes are used as the embedded pilot sequence. At the receiver, we perform cross-correlation-based channel estimation utilizing the unique autocorrelation properties of the pilot code. Comparing the simulation results, we find that our proposed channel estimation scheme provides good performance while keeping the signal peak and PAPR low. Topics of future research include adapting the proposed method for fractional Doppler, comprehensive analysis of potential code sets for a pilot code-based channel estimation method for MIMO-OTFS systems and exploring strategies for improving spectral efficiency.

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